Conic Sections — Sign and Denominator Errors
Answer Key

Algebra 2

Quick Answers

1. x = -4, y = 7, 6
2. x = 5, y = -2, 7
3. 5, 3
4. 3, -5, 4
5. 14
6. x = -3, y = 1, 4
7. -3, 3, 3
8. —

Curriculum

Level: Algebra 2
HSF-IF.A–HSF-IF.C – *problems 7, 8*
HSG-GPE.A.1–A.3 – *problems 1, 2, 3, 4, 5, 6*
Difficulty 1–5 of 5 across 15 tagged checks.

Common wrong answers

1. *If they got 36: read the number on the right as the radius instead of taking its square root*
2. *If they got 49: read the number on the right as the radius instead of taking its square root*
3. *If they got 25: used the denominator itself as the semi-axis length instead of its square root*
3. *If they got 9: used the denominator itself as the semi-axis length instead of its square root*
4. *If they got -3: halved the x coefficient but copied its sign instead of flipping it*
4. *If they got 5: halved the y coefficient but copied its sign instead of flipping it*
4. *If they got 7.21: added the constant term to the right-hand side instead of subtracting it*
5. *If they got 4: took the major axis from the denominator under x squared because it is written first*
5. *If they got 7: stopped at the semi-axis instead of doubling it for the full axis length*
6. *If they got 16: used the denominator itself as the semi-axis length instead of its square root*
6. *If they got 2: picked the smaller denominator, so the shorter axis was called the major one*
7. *If they got 4: took the vertex distance from the denominator under the subtracted term*

1. Centre and radius of $(x+4)^2 + (y-7)^2 = 36$.

Step 1: the standard form is $(x-h)^2 + (y-k)^2 = r^2$, so h and k are the values that make each binomial zero. Set $x+4=0$ and $y-7=0$.
Step 2: $x=-4$ and $y=7$, so the centre is $\boxed{(-4,\ 7)}$. The printed $+4$ becomes a negative coordinate; the printed -7 stays positive.
Step 3: the right-hand side is r^2, not r. From $r^2 = 36$ take the positive square root: $r = \boxed{6}$.
Watch for: a radius reported as 36. That is r^2; a circle of radius 36 would be six times too big.

2. Find and fix: Dev's $(x-5)^2 + (y+2)^2 = 49$.

Step 1: name the first error. Dev flipped both signs the wrong way. The form subtracts the centre coordinates, so $(x-5)$ already means $h = +5$ and $(y+2)$ means $k = -2$. Dev reversed a subtraction that was not there and left the addition alone.
Step 2: set each binomial to zero: $x - 5 = 0$ gives $x = 5$; $y + 2 = 0$ gives $y = -2$. The correct centre is $\boxed{(5,\ -2)}$.
Step 3: name the second error. Dev copied the right-hand side as the radius. It is r^2, so $r = \sqrt{49}$, and the correct radius is $\boxed{7}$.
Full-credit answer says: "the centre signs were reversed" and "the right side is r^2, so the radius is its square root."

3. Semi-axes of $\dfrac{x^2}{25} + \dfrac{y^2}{9} = 1$.

Step 1: in $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ each denominator is a *square*, so each semi-axis is the square root of its denominator — never the denominator itself.
Step 2: under x^2 the denominator is 25, so that semi-axis is $\sqrt{25} = \boxed{5}$.
Step 3: under y^2 the denominator is 9, so that semi-axis is $\sqrt{9} = \boxed{3}$.
Step 4: the larger of the two, 5, lies along the x-axis, so this ellipse has a horizontal major axis with semi-major length 5 and semi-minor length 3.
Watch for: semi-axes reported as 25 and 9 — the denominators themselves. Sketching the ellipse catches it instantly: it would run off any reasonable set of axes.

4. General form: $x^2 + y^2 - 6x + 10y + 18 = 0$.

Step 1: group and move the constant across. $(x^2 - 6x) + (y^2 + 10y) = -18$. The constant changes sign when it crosses the equals sign — this is the step where a stray $+18$ on the right does its damage.
Step 2: complete the square in x. Half of the x coefficient is $\dfrac{-6}{2} = -3$, and the centre coordinate is its opposite, so $h = \boxed{3}$. Add $(-3)^2 = 9$ to both sides.
Step 3: complete the square in y. Half of the y coefficient is $\dfrac{10}{2} = 5$, and the centre coordinate is its opposite, so $k = \boxed{-5}$. Add $5^2 = 25$ to both sides.
Step 4: collect the right-hand side: $-18 + 9 + 25 = 16$, so $r^2 = 16$ and $r = \boxed{4}$. The circle is $(x-3)^2 + (y+5)^2 = 16$.
Watch for: adding the constant instead of subtracting it gives $18 + 9 + 25 = 52$ and a radius near 7.21 — a circle almost twice the right size. Also watch the sign flip in steps 2 and 3: half of -6 is -3, but the centre coordinate is $+3$.

5. Find and fix: Rae's $\dfrac{x^2}{4} + \dfrac{y^2}{49} = 1$.

Step 1: name the rule Rae needed. The major axis lies along whichever variable carries the *larger* denominator, not whichever term is written first. Here $49 > 4$ and 49 sits under y^2, so the major axis is **vertical**, not horizontal.
Step 2: take the square root, because the denominator is a square: the semi-major axis is $\sqrt{49} = 7$.
Step 3: the full major axis is twice the semi-axis, so its length is $2 \times 7 = \boxed{14}$, running vertically.
Full-credit answer says: "the larger denominator decides the major axis," and shows both missing steps — the square root and the doubling.
Watch for: the reported 4. That is the denominator under x^2 used raw — wrong axis, no square root and no doubling, three errors in one number. A student who answers 7 has the right axis but stopped at the semi-axis.

6. Shifted ellipse $\dfrac{(x+3)^2}{16} + \dfrac{(y-1)^2}{4} = 1$.

Step 1: the centre is where both squared binomials vanish. Set $x + 3 = 0$ and $y - 1 = 0$.
Step 2: $x = -3$ and $y = 1$, so the centre is $\boxed{(-3,\ 1)}$. A shift does not change the denominators — it only moves the whole ellipse.
Step 3: the larger denominator is 16, under the x term, so the major axis is horizontal and the semi-major axis is $\sqrt{16} = \boxed{4}$.
Watch for: a semi-axis of 16 (the denominator uncorrected) or of 2 (the square root of the *smaller* denominator, which is the semi-minor axis). Both are the same reading error at different stages.

7. Find and fix: Sam's $\dfrac{y^2}{9} - \dfrac{x^2}{16} = 1$.

Step 1: name the deciding feature. In a hyperbola the two squared terms are *subtracted*, and the branches open along the variable whose term is **positive**. Here the positive term is the y term, so the branches open up and down — not left and right. The subtracted term never touches the curve.
Step 2: find the vertices by setting $x = 0$, which is where the curve meets its own axis: $\dfrac{y^2}{9} = 1$, so $y^2 = 9$ and $y = \boxed{-3}$ or $y = \boxed{3}$.
Step 3: the vertices are therefore $(0,\ -3)$ and $(0,\ 3)$ — three units above and below the centre, on the y-axis.
Full-credit answer says: "the positive term decides the direction, and its denominator gives the vertex distance."
Watch for: Sam's 4. That is $\sqrt{16}$, the denominator under the subtracted term, so it is the wrong number *and* on the wrong axis. It would place vertices where the hyperbola has no points at all.

8. Show what you know: one centre, three conics.

Open task — flagged for manual review. A model answer is below; accept any equations that meet the three conditions.

Model, all centred at $(2,\ -5)$:

- circle: $(x-2)^2+(y+5)^2=36$;
- ellipse, vertical major axis: $\dfrac{(x-2)^2}{9}+\dfrac{(y+5)^2}{25}=1$;
- hyperbola, opening left and right: $\dfrac{(x-2)^2}{9}-\dfrac{(y+5)^2}{25}=1$.

(d) Model explanation: "Both squared terms are added in the first two and subtracted in the third, so the third is the hyperbola. Between the first two, the circle has a single number on the right and no denominators, while the ellipse has two different denominators. The centre is read the same way in all three: set each binomial to zero, so $x-2=0$ and $y+5=0$ give the centre $(2,\ -5)$ — the printed sign is the opposite of the coordinate."

What a full-credit answer must contain: all three equations sharing one non-origin centre; the vertical major axis shown by the larger denominator sitting under the y term; the hyperbola opening left and right shown by the x term carrying the plus sign; and an explanation that mentions both the sign rule for the centre and the fact that denominators are squares.